

Aggregating Judgments when People Receive Advice from an Artificial Intelligence System

Preregistration

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Overview

In many managerial settings, good decision making depends on obtaining an accurate forecast of an uncertain variable of interest. Useful knowledge about the variable may be accessible to both human experts and AI models, where each may have their own relative advantages in forming an accurate forecast. To make use of both sources of information, AI advice can be provided to human experts who can assess how much weight it should be given when updating their prior forecasts. Since useful information may be held by different individuals, combining forecasts from multiple experts can also boost the accuracy of the aggregate forecast. However, while providing AI advice to humans can be helpful at an individual level, it may also induce correlations in judgment errors that hamper efficient combination of information at the aggregate level. Using a stylized Bayesian model of information held by human experts and the algorithm, we have derived a new procedure for aggregating judgments when humans receive AI advice. The method uses individual responses to estimate how much weight to put on human versus AI forecasts and forms an aggregate forecast by taking the final average human forecast and adjusting it back toward the initial average human forecast.

The purpose of this online experiment is to evaluate the performance of this proposed judgment aggregation procedure and to test whether it can outperform existing alternatives.

Our primary research questions are:

- (1) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the average initial human forecast?
- (2) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the average final human forecast?
- (3) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the AI advice?

Details of our proposed judgment aggregation method:

In our setting, a crowd of J judges each begins by providing their own, individual estimate of the variable of interest, before receiving any advice. The true value of the unknown variable of interest is θ . We denote the initial estimate given by judge j by f_j^0 . Next, all judges are provided with the same algorithmic advice, denoted by f^a , and asked to update their estimate of the variable of interest. We denote the final estimate given by judge j by f_j^1 . We denote the means of the initial and final estimates over all J judges as $\bar{f}^0$ and $\bar{f}^1$, respectively.

After gathering all of the responses from all judges, we will estimate the weight on advice used by the judges according to equation (5) below:

$$(f_j^1 - f_j^0) = w(f^a - f_j^0) + \varepsilon_j^1. \quad (5)$$

Of note, in this specification we regress the change in forecast $f_j^1 - f_j^0$ on $f^a - f_j^0$, where the intercept is fixed equal to zero. The coefficient estimate $\hat{w}$ has standard error $\hat{\sigma}_{\hat{w}}$ and $J - 1$ degrees of freedom.

Once this regression has been estimated, we form our proposed aggregate forecast according to:

$$\hat{\theta} = \bar{f}^1 + \hat{\gamma}(\bar{f}^0 - \bar{f}^1), \quad (6)$$

where the pivoting factor $\hat{\gamma} = \frac{1}{T_{\hat{w}, \hat{\sigma}_{\hat{w}}, J-1}(1) - T_{\hat{w}, \hat{\sigma}_{\hat{w}}, J-1}(\underline{w})} \int_{\underline{w}}^1 \left(1 - \frac{\frac{w}{1-w} - \frac{J-1}{J} \frac{\sigma_0^2}{\sigma_x^2}}{w \left(1 + \frac{w}{1-w} - \frac{J-1}{J} \frac{\sigma_0^2}{\sigma_x^2} \right)} \right) t_{\hat{w}, \hat{\sigma}_{\hat{w}}, J-1}(w) dw$,
 $\underline{w} = \frac{\frac{J-1}{J} \frac{\sigma_0^2}{\sigma_x^2}}{1 + \frac{J-1}{J} \frac{\sigma_0^2}{\sigma_x^2}}$, and $(\hat{w}, \hat{\sigma}_{\hat{w}})$ are the coefficient estimate and its standard error from the ordinary least-squares regression specified in equation (5).

We calculate $\sigma_0^2 = \frac{1}{J-1} \sum_{j=1}^J (f_j^0 - \bar{f}^0)^2$ as the population variance of the initial human estimates. To calculate the AI's forecast accuracy, we use the out-of-sample accuracy of the AI's prediction on a different dataset containing N questions following $\sigma_x^2 = \frac{1}{N} \sum_{i=N}^N (f_i^a - \theta_i)^2$, where f_i^a is the AI's prediction for task i .

$T_{m,s,n-1}(x)$ and $t_{m,s,n-1}(x)$ are the CDF and PDF, respectively of a Student's t -distribution with location m , scale s , and degrees of freedom $n - 1$, evaluated at the point x .

Procedure

In this experiment, participants will estimate the age (in years) of an anonymous person based on a photograph. For each photograph, we will first ask the participants to enter their initial estimate. Subsequently, the next screen will show the participant an estimate generated by the AI advisor,

along with a paragraph of text containing the AI's explanation for its estimate. After received the AI advice and reading its explanation, the participant will be asked to provide a final estimate of the age of the person in the photograph. The bonus earned by a participant will depend on the accuracy of both of their estimates – without and with AI advice.

Prior to the main estimation task, there will be two examples to check that the participant understands the bonus payment formula. Next, the participants will be led through a set of three example questions to help them understand the task: how we are eliciting estimates and how their responses are incentivized so that it is in their interest to provide as accurate an estimate as possible on both instances (before and after AI advice). To familiarize themselves with the AI, the participants will also be given some information about the effectiveness of the AI and a measure of its prediction accuracy. The three practice questions will be drawn from a different set of photographs which are not used in the main study. This is the same dataset using which we will be calculating σ_x^2 as defined above.

The bonus earned from two estimates on each question are calculated independently of each other. Participants will receive a fixed payment of \$1 plus an additional bonus payment of between \$0 and \$0.10 for each estimate they provide. For every year that the estimate is off by, 1 cent will be deducted from the maximum bonus amount of \$0.10. If this formula gives a negative value, zero bonus will be paid for that question.

Every participant will be providing two estimates for each of 10 photographs (excluding the three photographs used in the example questions). This means a total of 20 estimates from each participant, with a corresponding total bonus earning potential of \$2.

At the end of the experiment, participants will answer a short, optional demographic questionnaire asking about their age and gender.

Planned Sample Size

We plan to collect data from 1100 participants, who will each receive a random subset of 10 photographs that are evenly drawn from a data set of 200 total photographs. This should yield approximately 55 pairs of participant judgments for each of the 200 photographs.

Exclusion Criteria

Participants' data will only be excluded if they:

- Complete the survey in an implausibly short amount of time (less than 3 minutes total).
- Fail to complete all of the required questions on the survey (all questions except for the optional demographic questions at the end).

Hypotheses

Based on the responses received for each question, we will first run the regression in Equation 5 on individual responses and then calculate our model's estimate of the uncertain parameter using Equation 6.

Our hypotheses (corresponding to the research questions on page 1) are that:

H1. *Yes, the final estimate of the unknown quantity derived using our procedure after observing participants' judgment aggregation behavior will be more accurate than the average initial human forecast.*

H2. *Yes, the final estimate of the unknown quantity derived using our procedure after observing participants' judgment aggregation behavior will be more accurate than the than the average updated human forecast.*

H3. *Yes, the final estimate of the unknown quantity derived using our procedure after observing participants' judgment aggregation behavior will be more accurate than the AI advice.*

Planned Hypothesis Tests

To compare the accuracy across the different judgment aggregation methods, we will first compute the forecast errors of each method over a set of 200 items (i.e., 200 different photographs of people's faces), where the answer for question i (the true age of the person in the photograph) is denoted by θ_i .

1. The calculations described in equations (5) and (6) above will be used to calculate the forecast $\hat{\theta}$ of our proposed judgment aggregation method. The forecast error of our method will be calculated as:

$$\hat{e} = \theta - \hat{\theta}.$$

2. The forecast error of the average initial human forecast will be calculated as follows:

$$e^0 = \theta - \bar{f}^0 = \theta - \frac{1}{J} \sum_{j=1}^J f_j^0.$$

3. The forecast error of the average initial human forecast will be calculated as follows:

$$e^1 = \theta - \bar{f}^1 = \theta - \frac{1}{J} \sum_{j=1}^J f_j^1.$$

4. The forecast error of the AI advice will be calculated as follows:

$$e^a = \theta - f^a.$$

We will compare the pairwise differences in absolute errors across the 200 photographs to test hypotheses H1, H2, H3 respectively. We will use two-sided paired t-tests to make each comparison and will set $\alpha = 0.05$ as our threshold for determining statistical significance for each test.

Images From the Experimental Interface

The following screen shows the initial judgment elicitation step. In this step, the only stimulus provided is a photograph of the person whose age is to be estimated.



What is your best guess of the **age (in years)** of the person in the above photograph?

Type your answer into the text box below.



The following screen shows the updated judgment elicitation step after receiving the AI advice. In this step, in addition to the photograph of the same person, the AI advice is displayed along with a two- or three-sentence-long paragraph of text which explains the AI reasoning underlying its estimate.



Our AI system estimated that the person in the photograph is **53 years old**, offering the following explanation for its prediction:

The age estimation is based on the combination of his hair, which shows signs of thinning and receding, and his skin, which displays some wrinkles and reduced elasticity. The facial features still retain a certain level of youthfulness, yet show the beginning signs of middle age, which supports this age bracket estimation.

In consideration of this advice, what is your best guess of the age of the person in the above photograph?

Type your answer into the text box below.

